# Introduction to Probability and Applied Statistics

Week 8: Linear Regression I

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# Roadmap

Relating two variables

Where we are

The difference in means is a regression

From conditional means to a line

Least squares

Interpreting the line

How well does the line fit?

Wrapping up

# Why relate two variables?

- Do cities with more renters have lower council-election turnout?
- Do women in a legislature speak for less time in committee hearings than men do?
- Do older citizens vote at higher rates than younger ones?
- Do women rate Kamala Harris more warmly than men do?
- How do ratings of Harris change from strong Democrats to strong Republicans?

Each asks: how does the average of $Y$ differ across values of $X$?

# Outcome and explanatory variables

*X*

The variable we use to describe or predict *Y*.

explanatory variable
independent variable
covariate
regressor

*Y*

The variable we want to describe or predict.

outcome
dependent variable
response variable
regressand

# Where we are going

The rest of the course is about one tool: **linear regression**.

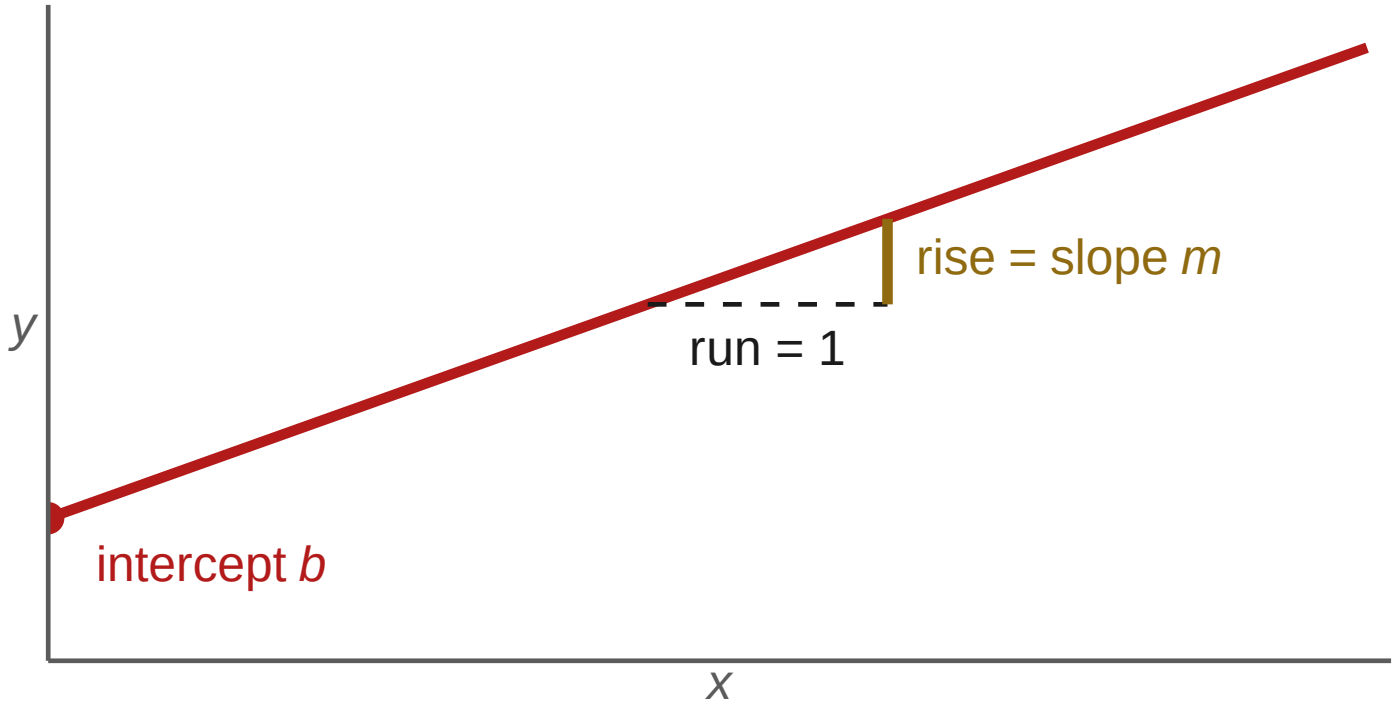
- Fitting a line to two variables, interpreting it, measuring its fit
- How the fitted line varies from sample to sample
- More than one $X$; group indicators, interactions, curves
- Tests and confidence intervals for regression
- Presenting results; regression and causal questions

# The best guess about *Y*

For any value of ***X***, what is the best guess about ***Y***?

The simplest way to relate two variables is a line:

$$y = mx + b$$



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# The mean is the best constant prediction

**THE MEAN AS A PREDICTION**

Predict the same number $c$ for every unit. The prediction with the smallest total squared error is the sample mean:

$$\bar{Y} = \arg\min_{c} \sum_{i} (Y_i - c)^2, \text{ uniquely.}$$

# The conditional mean

**CONDITIONAL MEAN**

The mean of $Y$ among the units with $X = x$:

$$\bar{Y}_{X=x} = \frac{\sum_i Y_i \mathbf{1}\{X_i = x\}}{\sum_i \mathbf{1}\{X_i = x\}}.$$

Read as a function of $x$, $\bar{Y}_{X=x}$ is the **conditional mean function**.

# The difference between two groups

**DIFFERENCE IN MEANS**

For two groups $A$ and $B$, the difference in average $Y$ between $A$ and $B$ is

$$\bar{Y}_A - \bar{Y}_B.$$

# Covariance and correlation

## SAMPLE COVARIANCE AND CORRELATION

$$\widehat{\mathrm{Cov}}(X, Y) = \frac{1}{n-1} \sum_i (X_i - \bar{X})(Y_i - \bar{Y}), \qquad \hat{r} = \frac{\widehat{\mathrm{Cov}}(X, Y)}{s_X \, s_Y}.$$

$s_X$ and $s_Y$ are the sample standard deviations, and $s_X^2 = \widehat{\mathrm{Cov}}(X, X)$.

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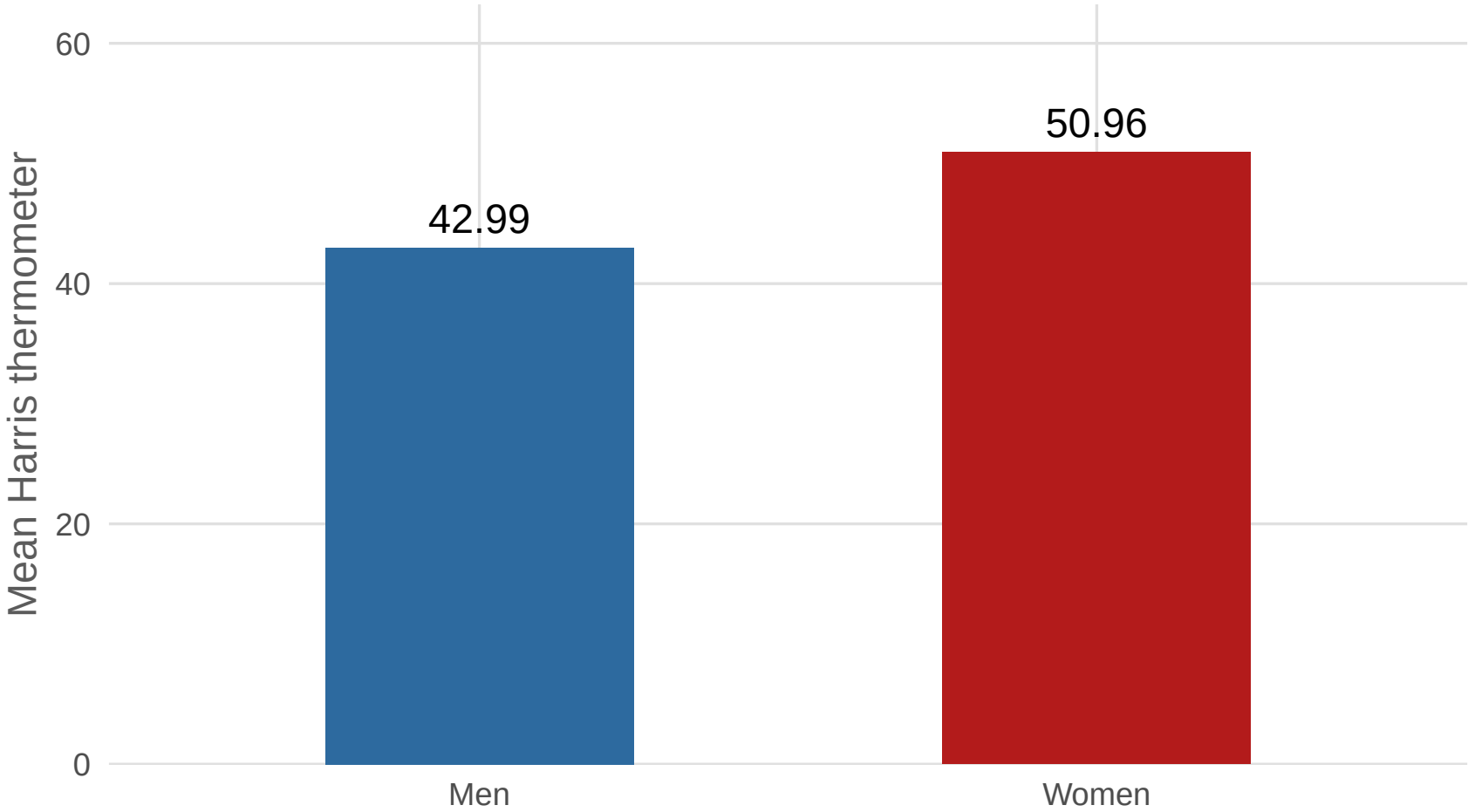
Wrapping up

# Two survey items

| Variable | ANES 2024 item | Values |
|---|---|---|
| `harris` | Feeling thermometer: Kamala Harris (V241156) | 0–100 |
| `female` | "What is your sex?" (V241550) | 1 = female, 0 = male |

Pre-election wave, unweighted, $n = 5{,}455$ respondents who answered both.

# Two group means



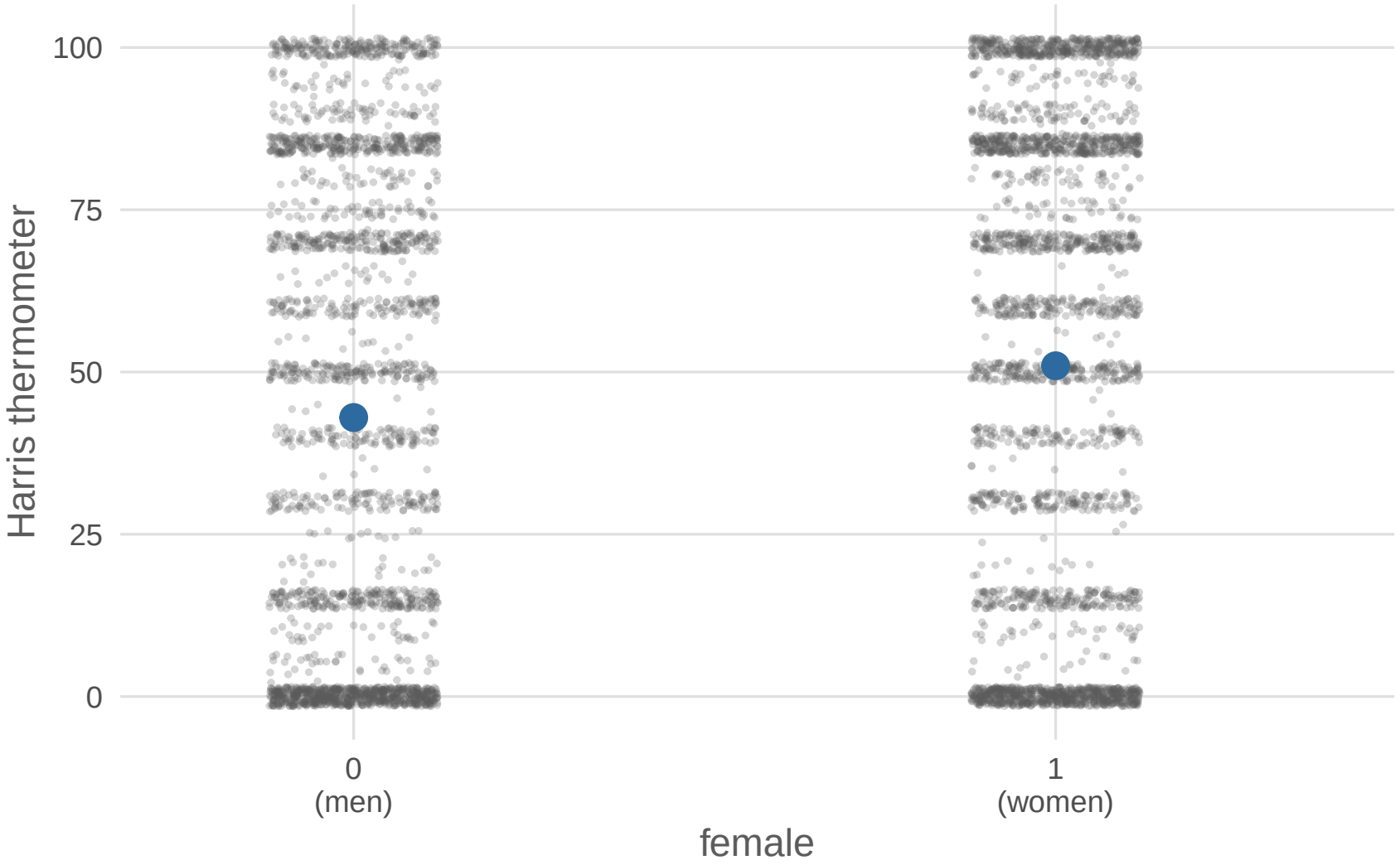

$$\bar{Y}_{\text{women}} - \bar{Y}_{\text{men}} = 50.96 - 42.99 = 7.97$$

# Sex as a 0/1 variable

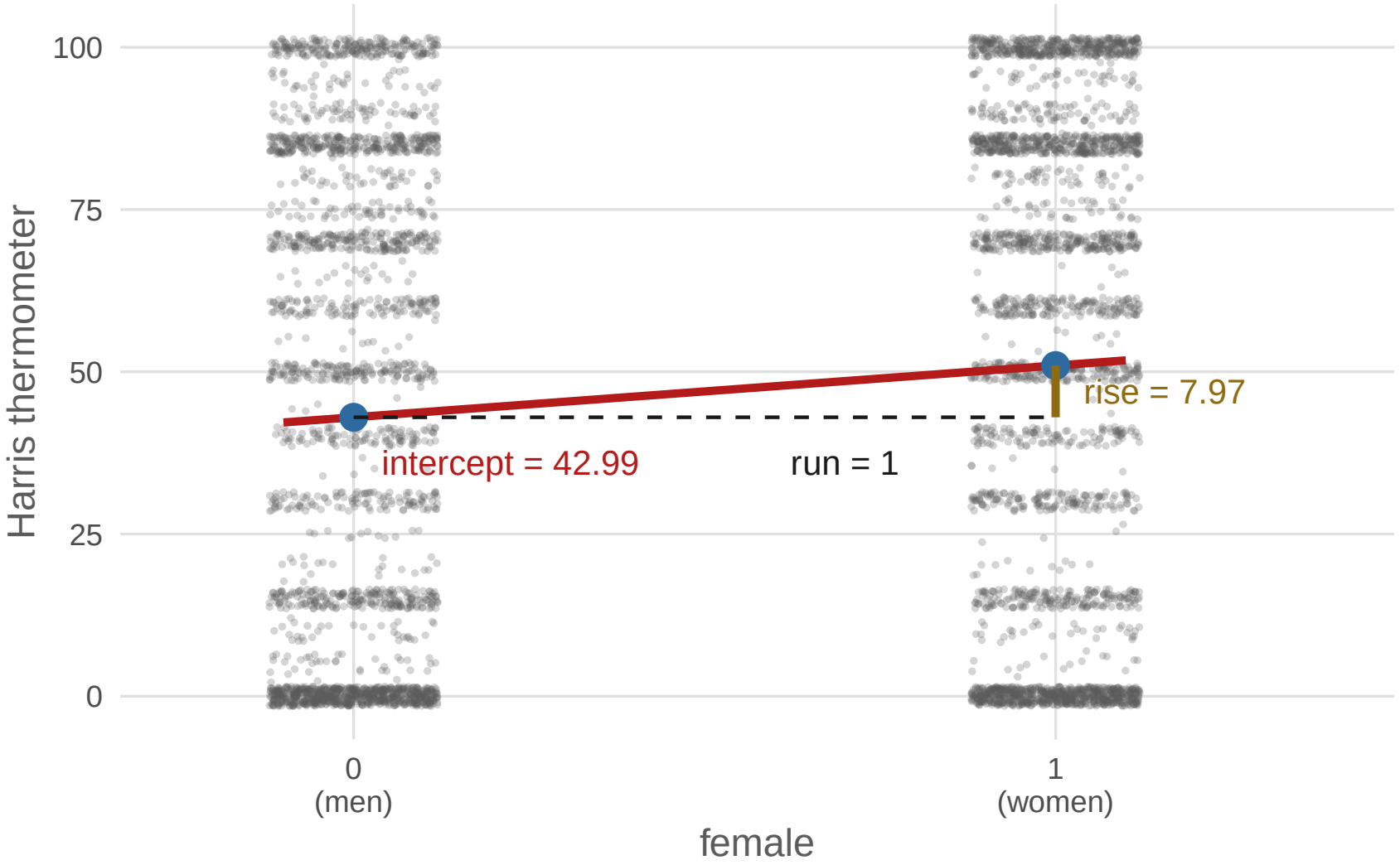
| id | harris | female |
|---|---|---|
| 140001 | 0 | 0 |
| 140002 | 50 | 1 |
| 140003 | 90 | 1 |
| 140004 | 50 | 0 |
| 140005 | 5 | 0 |
| 140006 | 85 | 1 |

A variable coded 1 for one group and 0 for the other is a **binary** or **dummy** variable.

# The ratings by group



# A line through the two means



# Asking R for the line

```
fit <- lm(harris ~ female, data = anes)
coef(fit)
```

```
(Intercept)      female
  42.989345    7.967519
```

# By hand and by `lm()`

**BY HAND**

$\bar{Y}_{X=0} = 42.99$

$\bar{Y}_{X=1} - \bar{Y}_{X=0} = 7.97$

**`lm()`**

intercept = 42.99

slope = 7.97

# Regression on a binary variable

**A LINE THROUGH TWO GROUP MEANS**

When $X$ takes only the values 0 and 1,

$$\text{intercept} = \bar{Y}_{X=0}, \qquad \text{slope} = \bar{Y}_{X=1} - \bar{Y}_{X=0}.$$

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# Party identification

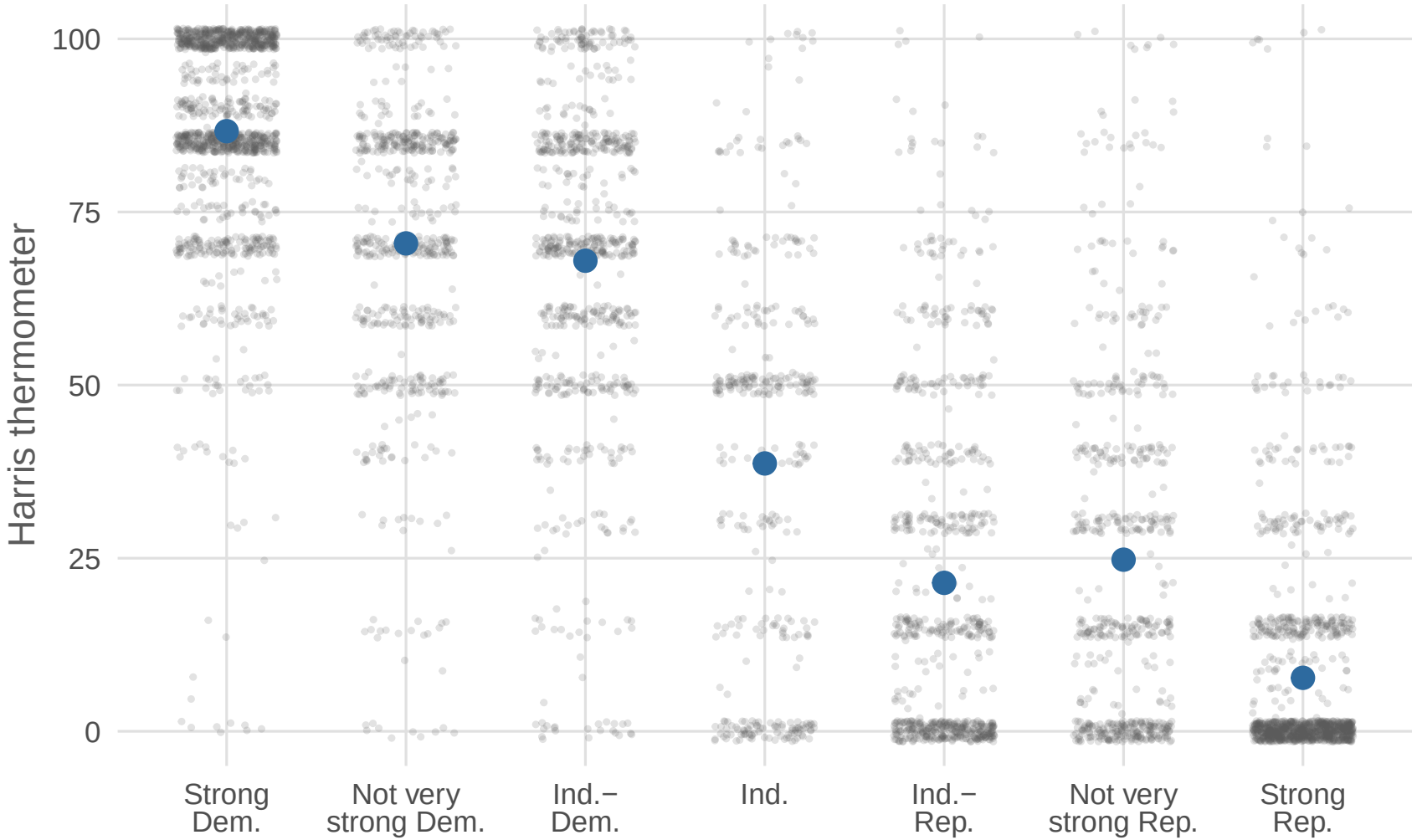
| `pid` | Party ID (V241227x) |
|---|---|
| 1 | Strong Democrat |
| 2 | Not very strong Democrat |
| 3 | Independent-Democrat |
| 4 | Independent |
| 5 | Independent-Republican |
| 6 | Not very strong Republican |
| 7 | Strong Republican |

| `harris` | Harris thermometer (V241156) |
|---|---|
| 0–49 | Unfavorable, cold |
| 50 | Neither warm nor cold |
| 51–100 | Favorable, warm |

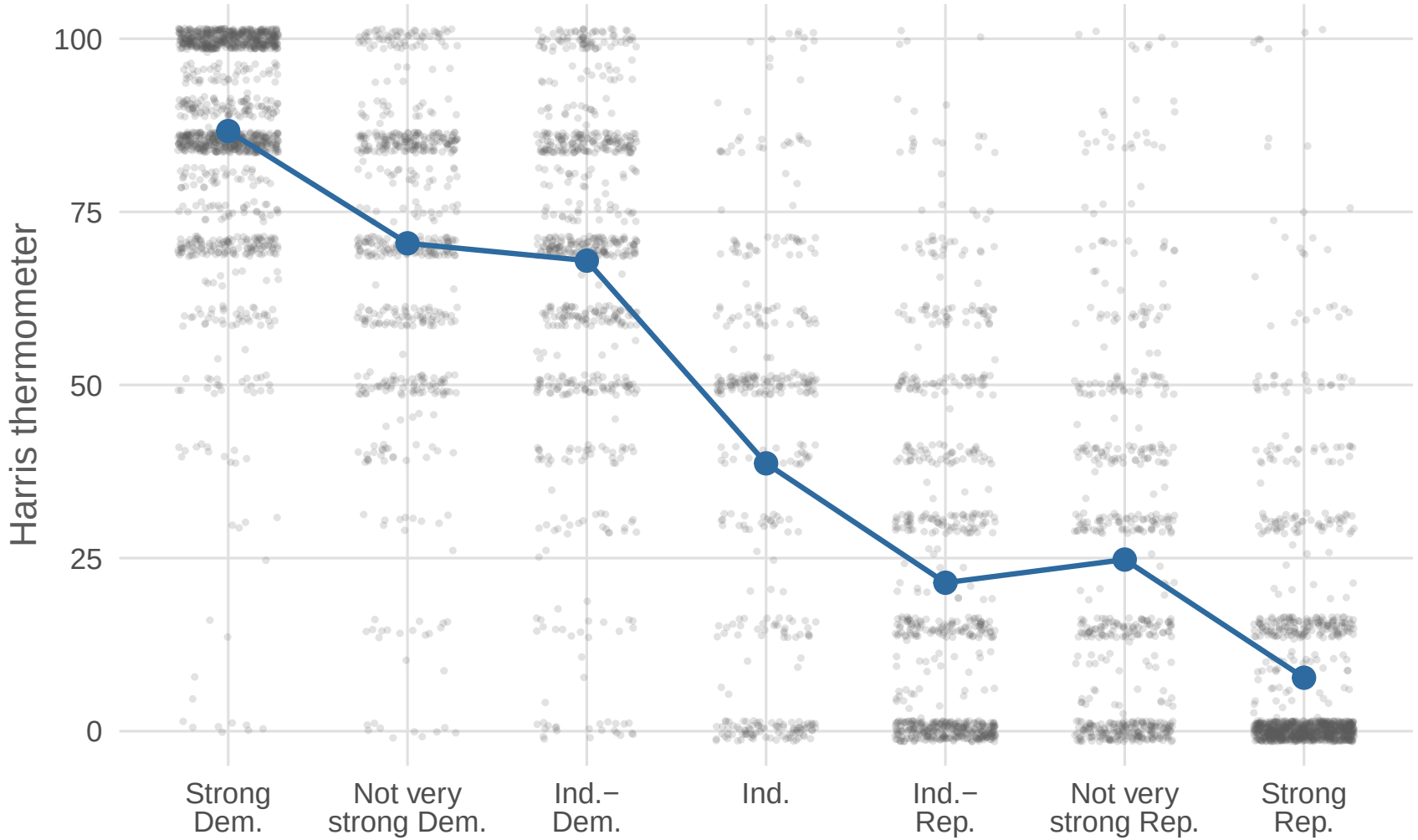
# Harris ratings by party ID

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| $\bar{Y}_{X=x}$ | 86.7 | 70.5 | 68.0 | 38.7 | 21.4 | 24.8 | 7.7 |
| $n$ | 1,312 | 613 | 713 | 377 | 714 | 575 | 1,163 |

# Every respondent, and the seven means



# The conditional mean function



# What should $f$ look like?

$$\mathbb{E}[Y \mid X = x] = f(x)$$

$\mathbb{E}[Y \mid X = x]$ is the population quantity that $\bar{Y}_{X=x}$ estimates.

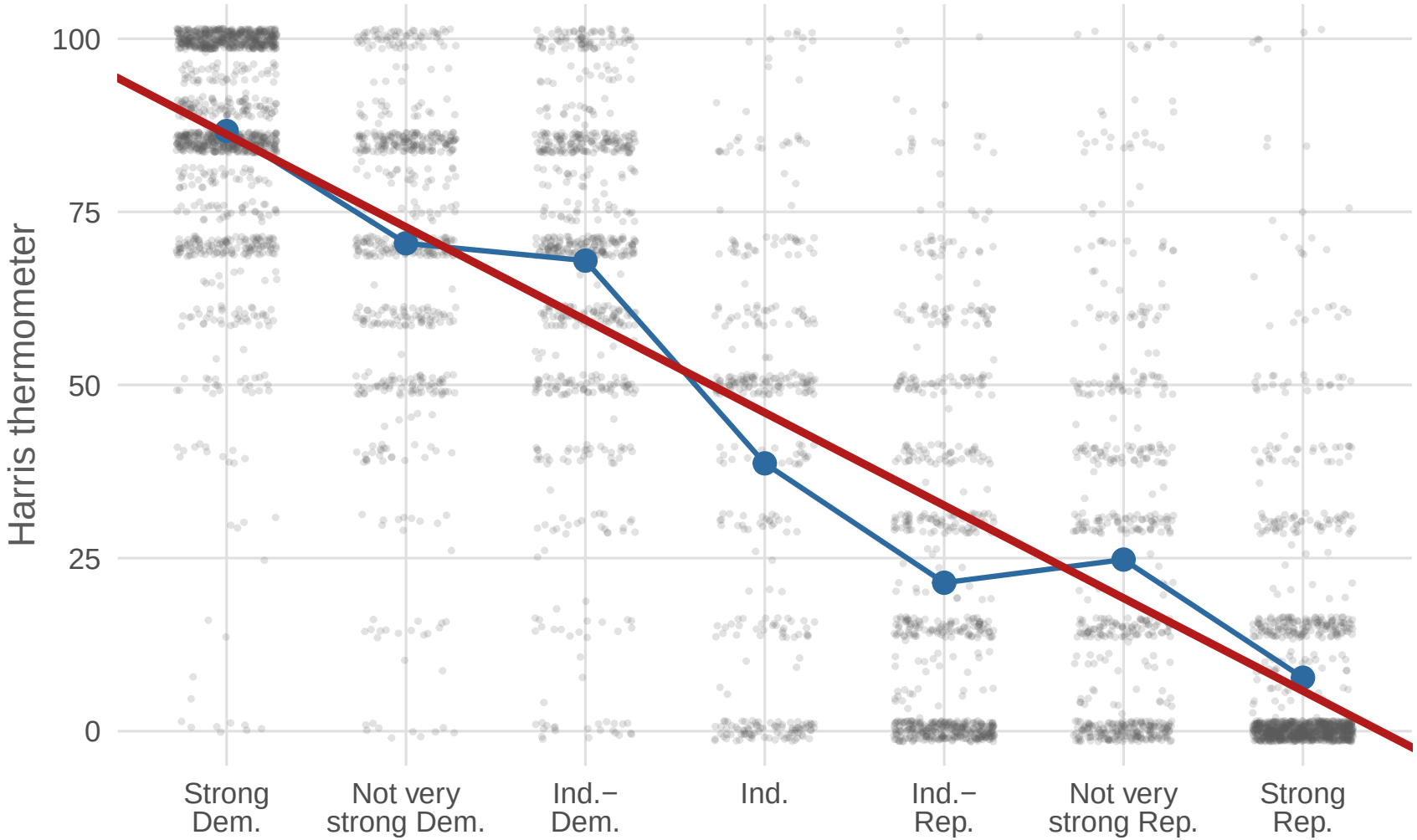
| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| $f(x)$ | $f(1)$ | $f(2)$ | $f(3)$ | $f(4)$ | $f(5)$ | $f(6)$ | $f(7)$ |
| $\bar{Y}_{X=x}$ | 86.7 | 70.5 | 68.0 | 38.7 | 21.4 | 24.8 | 7.7 |

**A LINEAR CONDITIONAL MEAN**

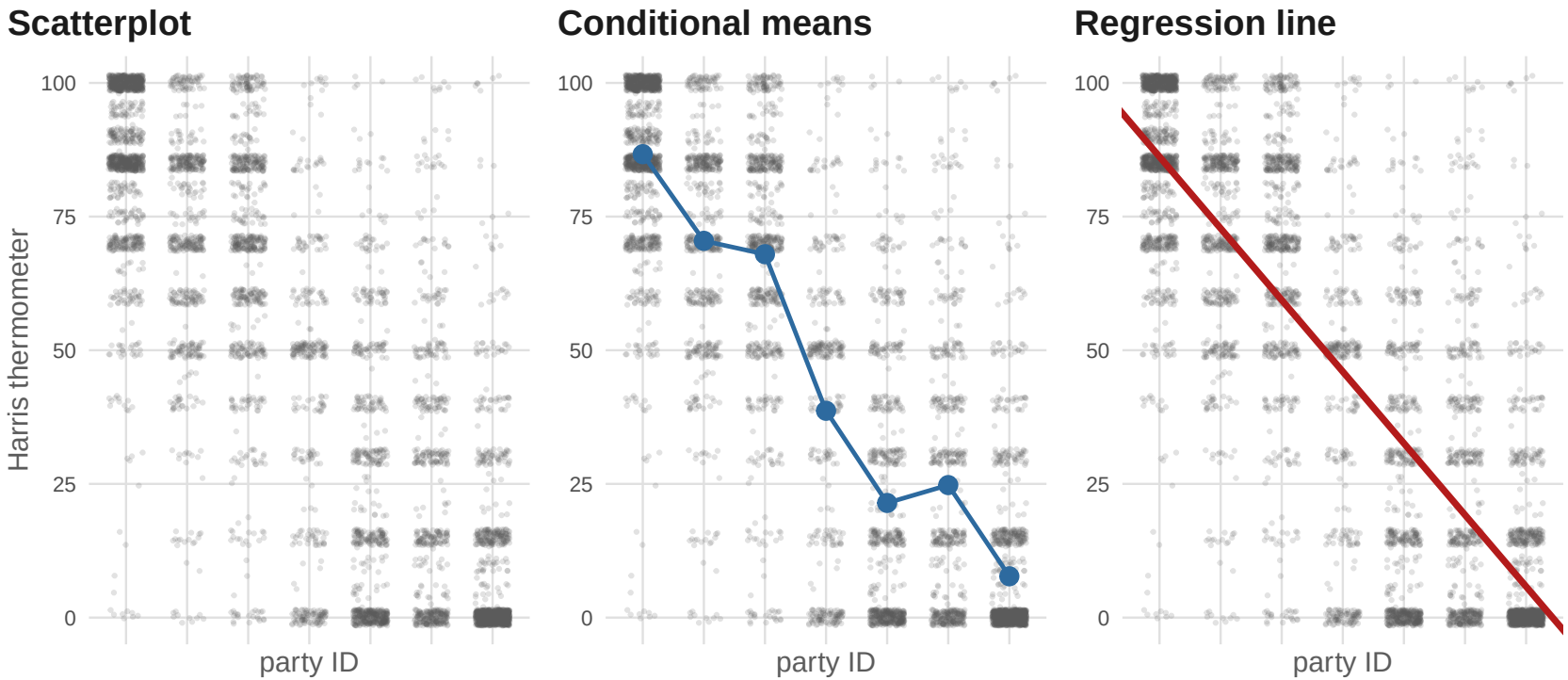
$$f(x) = \beta_0 + \beta_1 x$$

$\beta_0$ is the **intercept**, $\beta_1$ the **slope**. Together they are the **coefficients**.

# The line over the conditional means



# The line summarizes the conditional means

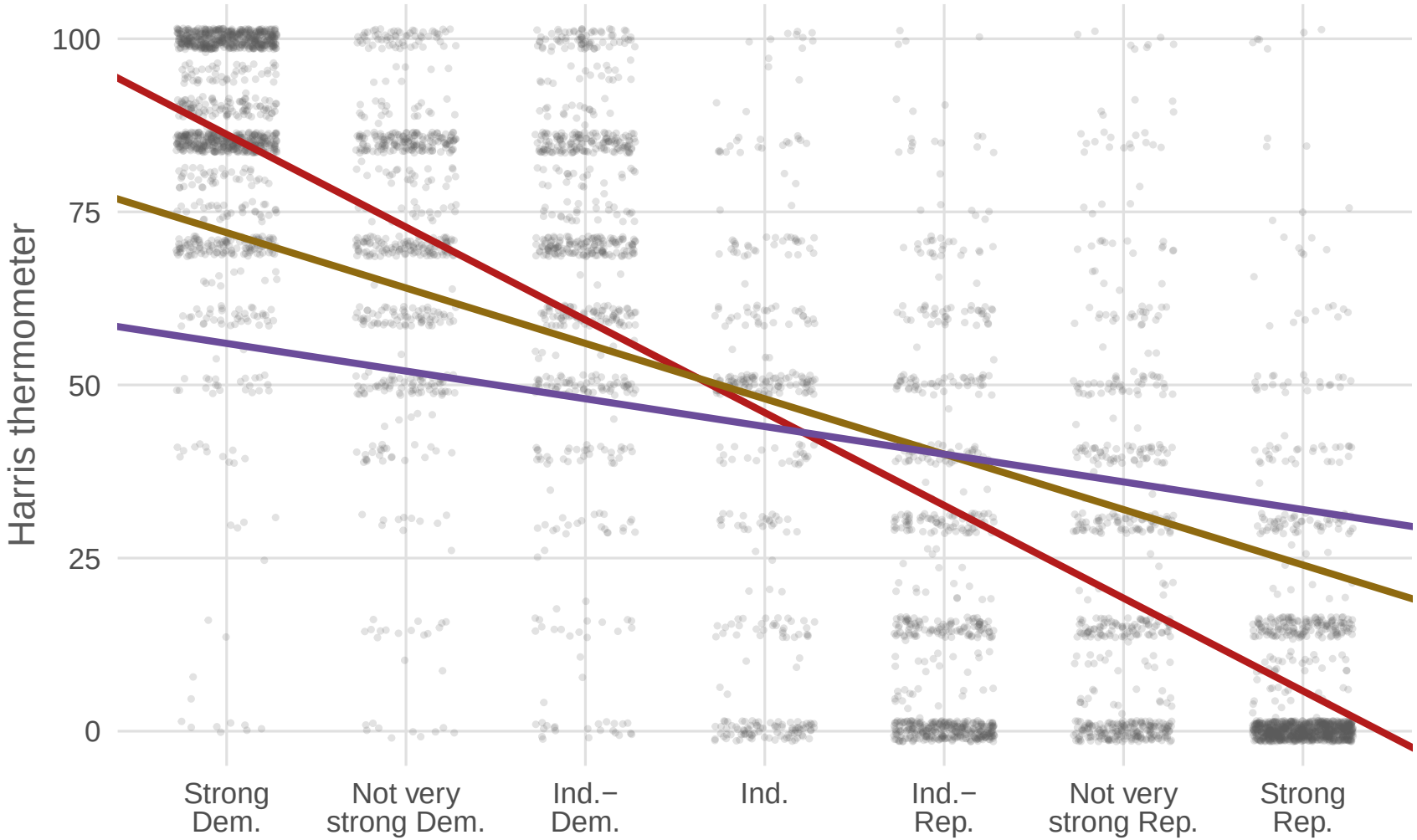


# Gailmard on the line

"A linear regression smooths out the jagged edges and peaks of the scatterplot and the conditional mean function. Neither the scatterplot as a whole nor a linear regression simplifying its patterns should be understood as the one true depiction of a relationship; these descriptive devices are more useful in conjunction with each other than as substitutes."

Gailmard, *Statistical Modeling and Inference for Social Science*, p. 44

# Which line?


Harris thermometer
100
75
50
25
0
Strong Dem.
Not very strong Dem.
Ind.–Dem.
Ind.
Ind.–Rep.
Not very strong Rep.
Strong Rep.

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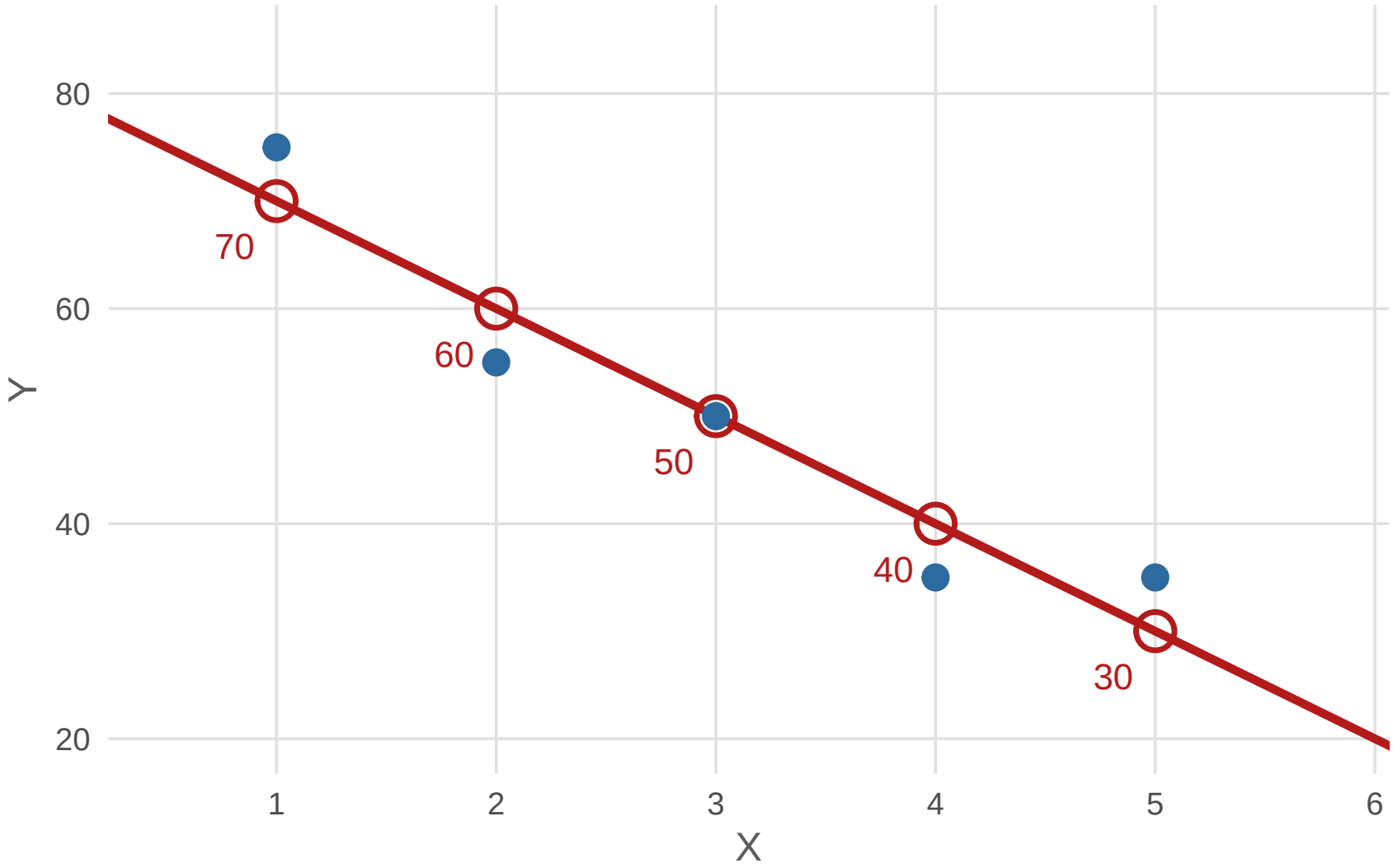
Least squares

Interpreting the line

How well does the line fit?

Wrapping up

# Fitted values



# Fitted values and residuals

**FITTED VALUE AND RESIDUAL**

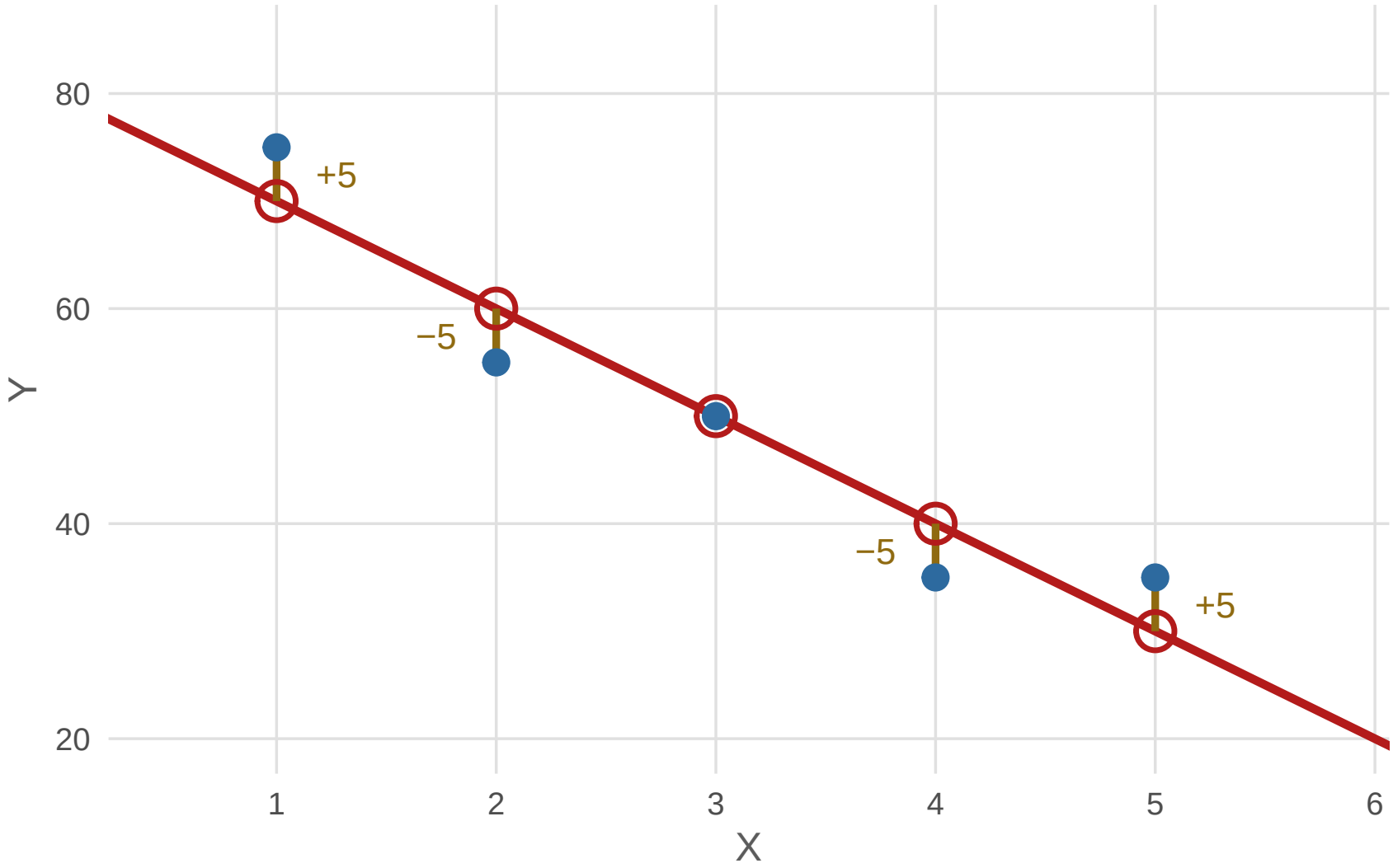
For a line with intercept $b_0$ and slope $b_1$, the **fitted value** for unit $i$ is

$$\hat{Y}_i = b_0 + b_1 X_i,$$

and the **residual** is the "miss,"

$$\hat{\varepsilon}_i = Y_i - \hat{Y}_i.$$

# Residuals



# The “best” line minimizes those distances

- We can’t just minimize $\sum_i \hat{\varepsilon}_i$ because positive and negative residuals cancel out.
- So we *square* the residuals.
- When $X$ takes at least two values, the squared criterion has exactly one best line.
- That line is our regression line.

# From a constant to a line

## BEST CONSTANT

$$\arg\min_{c} \sum_{i} (Y_i - c)^2$$

Answer: $\bar{Y}$

## BEST LINE

$$\arg\min_{b_0, b_1} \sum_{i} (Y_i - b_0 - b_1 X_i)^2$$

$\hat{Y}_i = b_0 + b_1 X_i$, so

$$Y_i - b_0 - b_1 X_i = Y_i - \hat{Y}_i = \hat{\varepsilon}_i,$$

the residual.

Answer: $\hat{\beta}_0$ and $\hat{\beta}_1$, the **ordinary least squares** (OLS) coefficients

# The OLS coefficients

The **sum of squared residuals** of a candidate line:

$$\text{SSR}(b_0, b_1) = \sum_{i=1}^{n} (Y_i - b_0 - b_1 X_i)^2 .$$

The OLS intercept and slope are the pair that makes SSR as small as possible:

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg\min_{b_0, b_1} \text{SSR}(b_0, b_1).$$

**BIVARIATE OLS**

$$\hat{\beta}_1 = \frac{\widehat{\text{Cov}}(X, Y)}{s_X^2}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

Every step of the derivation: `handout_ols_derivation.pdf`.

# Slope and correlation

$$\hat{\beta}_1 = \frac{\widehat{\text{Cov}}(X, Y)}{s_X^2}$$

The OLS slope.

$$= \frac{\widehat{\text{Cov}}(X, Y)}{s_X \cdot s_X} \cdot \frac{s_Y}{s_Y}$$

Write $s_X^2 = s_X \cdot s_X$ and multiply by $s_Y / s_Y = 1$.

$$= \frac{\widehat{\text{Cov}}(X, Y)}{s_X \, s_Y} \cdot \frac{s_Y}{s_X}$$

Swap the two denominators.

$$= \hat{r} \frac{s_Y}{s_X}$$

The definition of $\hat{r}$.

In the bivariate case, the slope is the correlation, rescaled from standard deviations into the units of $Y$ per unit of $X$.

# A toy example

| $i$ | $X_i$ | $Y_i$ | $X_i - \bar{X}$ | $Y_i - \bar{Y}$ | $(X_i - \bar{X})(Y_i - \bar{Y})$ | $(X_i - \bar{X})^2$ |
|---|---|---|---|---|---|---|
| 1 | 1 | 75 | −2 | 25 | −50 | 4 |
| 2 | 2 | 55 | −1 | 5 | −5 | 1 |
| 3 | 3 | 50 | 0 | 0 | 0 | 0 |
| 4 | 4 | 35 | 1 | −15 | −15 | 1 |
| 5 | 5 | 35 | 2 | −15 | −30 | 4 |
| Mean | 3 | 50 | | | | |
| Sum | | | 0 | 0 | **−100** | **10** |

$$\widehat{\text{Cov}}(X, Y) = \frac{-100}{5-1} = -25 \qquad s_X^2 = \frac{10}{5-1} = 2.5$$

# The toy example, solved

$$\widehat{\text{Cov}}(X, Y) = \frac{1}{n-1}\sum_i (X_i - \bar{X})(Y_i - \bar{Y}) = \frac{-100}{4} = -25$$

$$s_X^2 = \frac{1}{n-1}\sum_i (X_i - \bar{X})^2 = \frac{10}{4} = 2.5$$

$$\hat{\beta}_1 = \frac{\widehat{\text{Cov}}(X, Y)}{s_X^2} = \frac{-25}{2.5} = -10$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = 50 - (-10)(3) = 80$$

$$\hat{Y}_i = 80 - 10\, X_i$$

Another way: the covariance and the variance both divide by $n - 1$, so $n - 1$ cancels from the ratio:

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{-100}{10} = -10$$

# The toy example in R

```
toy <- data.frame(X = 1:5, Y = c(75, 55, 50, 35, 35))
coef(lm(Y ~ X, data = toy))
```

```
(Intercept)           X
         80         -10
```

# Harris therm and Party ID: the formula and `lm()`

$$\hat{\beta}_1 = \frac{\widehat{\text{Cov}}(X, Y)}{s_X^2}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

## THE FORMULA

```
d  <- na.omit(
  anes[c("pid", "harris")])
b1 <- cov(d$pid, d$harris) /
      var(d$pid)
b0 <- mean(d$harris) -
      b1 * mean(d$pid)
c(b0, b1)
```

```
[1]  99.591 -13.398
```

## `lm()`

```
fit <- lm(harris ~ pid,
          data = anes)
coef(fit)
```

```
(Intercept)         pid
     99.591     -13.398
```

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# The slope

**INTERPRETING $\hat{\beta}_1$**

$$\widehat{\texttt{harris}} = 99.59 - 13.40\,\texttt{pid}$$

On average, a one-unit increase in party ID is associated with a 13.4-point decrease in the Harris thermometer rating.

In general: on average, a one-unit increase in $X$ is associated with a $\hat{\beta}_1$-unit change in $Y$.

# Interpreting $\hat{\beta}_1$: check yourself

- What is a one-unit increase in party ID?

  One step on the 7-point scale, toward Republican: Strong Democrat (1) to Not very strong Democrat (2).

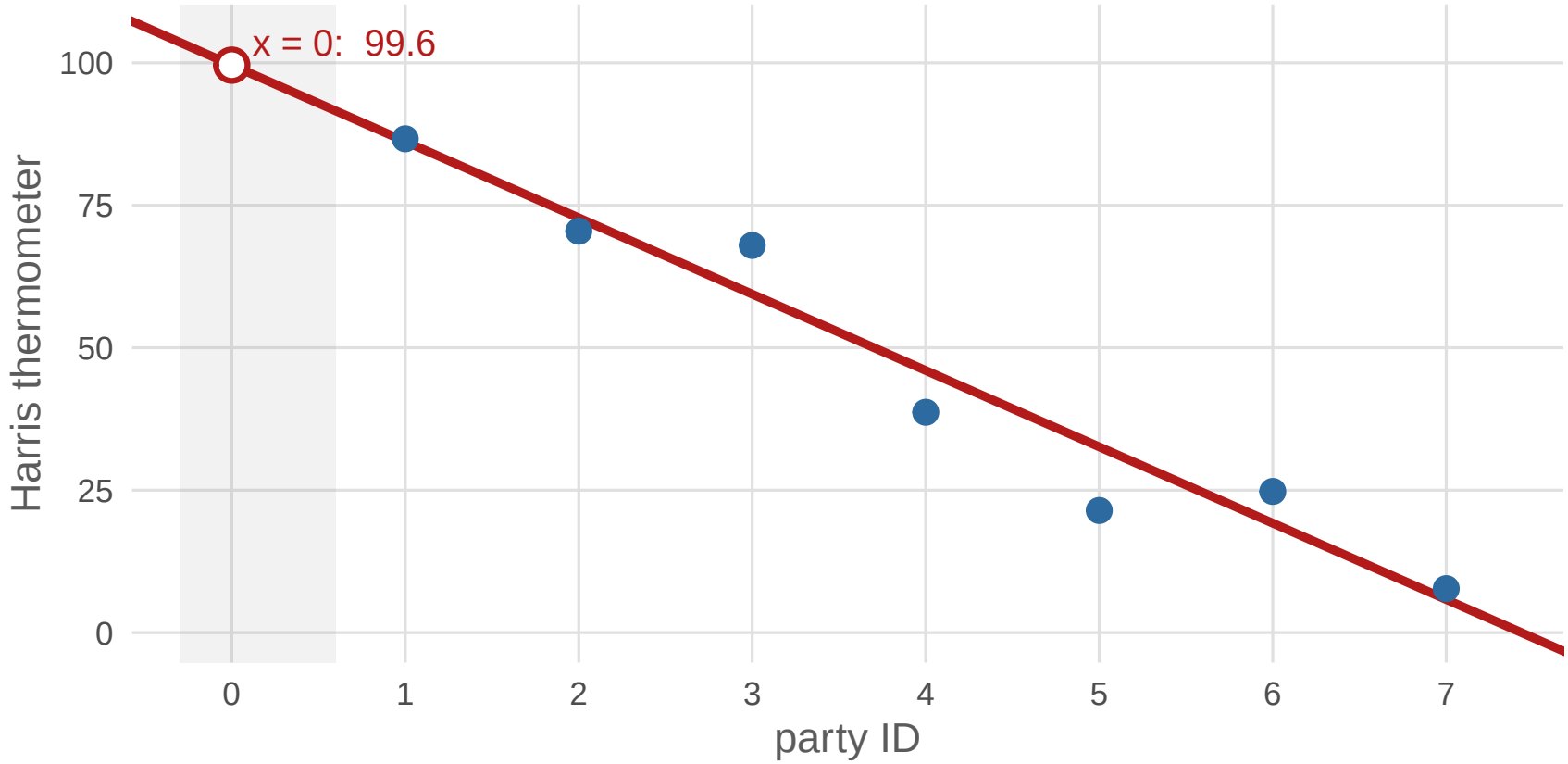
- A Strong Democrat (1) and a Strong Republican (7): how far apart are their predicted ratings?

  Six steps: $6 \times 13.4 = 80.4$ points.

- Does moving a respondent one step toward Republican lower their rating of Harris by 13.4 points?

  Careful! We say "is associated with", not "causes".

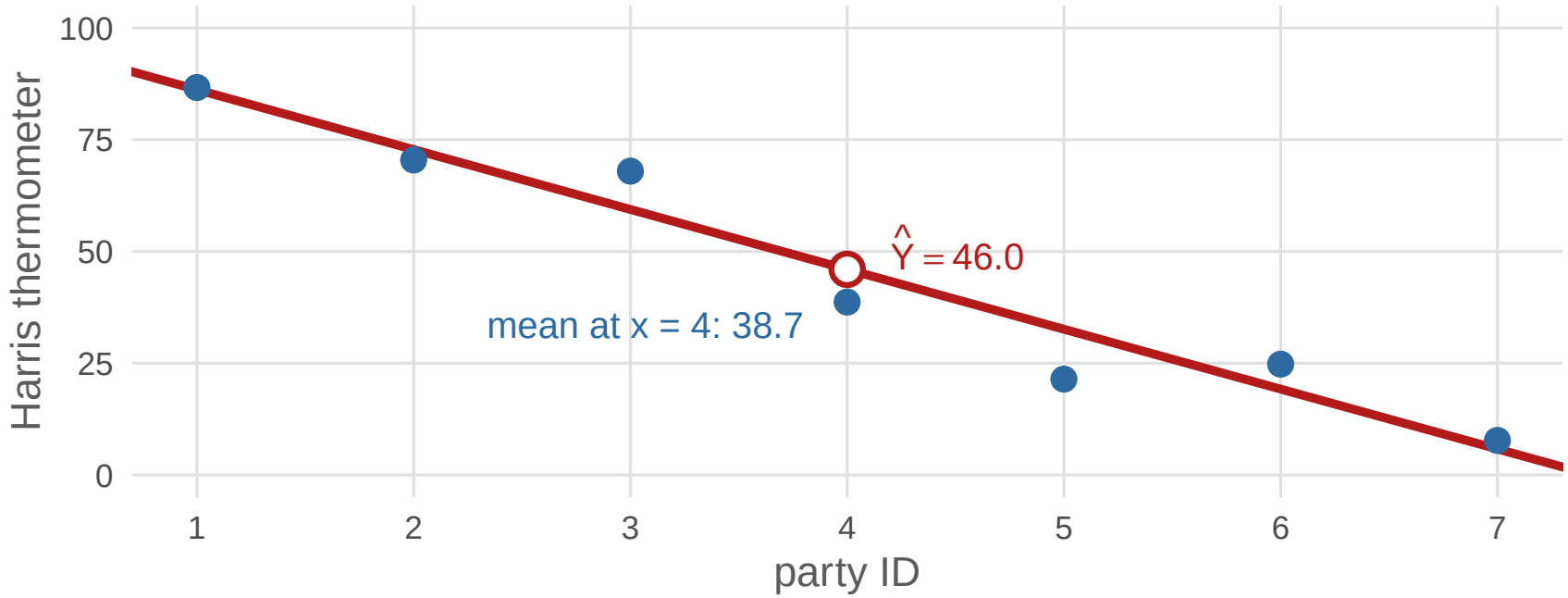
# The intercept




$\hat{\beta}_0$ = 99.6 is the predicted rating at party ID = 0. What does this mean?

# A prediction from the line

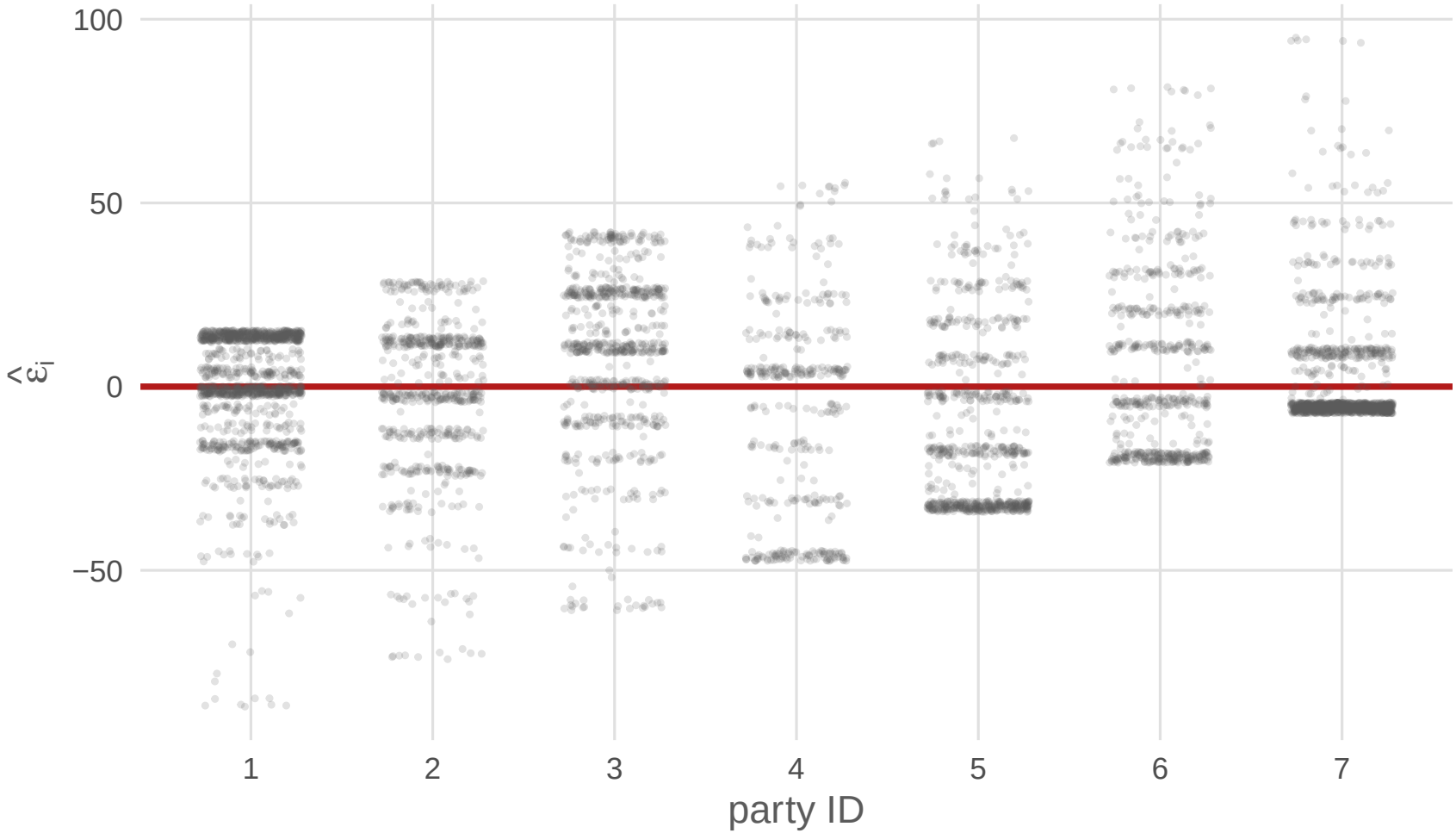
What is the predicted Harris rating for an Independent (`pid = 4`)?




$$\widehat{\texttt{harris}} = 99.59 - 13.40\,\texttt{pid}$$

$$\hat{Y} = 99.59 - 13.40 \times 4 = 46.0$$

# The residuals



# OLS residuals average to zero

**OLS RESIDUALS**

$$\sum_i \hat{\varepsilon}_i = 0 \qquad \text{and} \qquad \sum_i X_i \hat{\varepsilon}_i = 0.$$

The residuals average to zero, and they have no linear relationship with $X$. The line has "used up" all the linear information in $X$.

# The line passes through $(\bar{X}, \bar{Y})$

$$\hat{\beta}_0 + \hat{\beta}_1 \bar{X}$$

The line at $X = \bar{X}$.

$$= (\bar{Y} - \hat{\beta}_1 \bar{X}) + \hat{\beta}_1 \bar{X}$$

Substitute $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$.

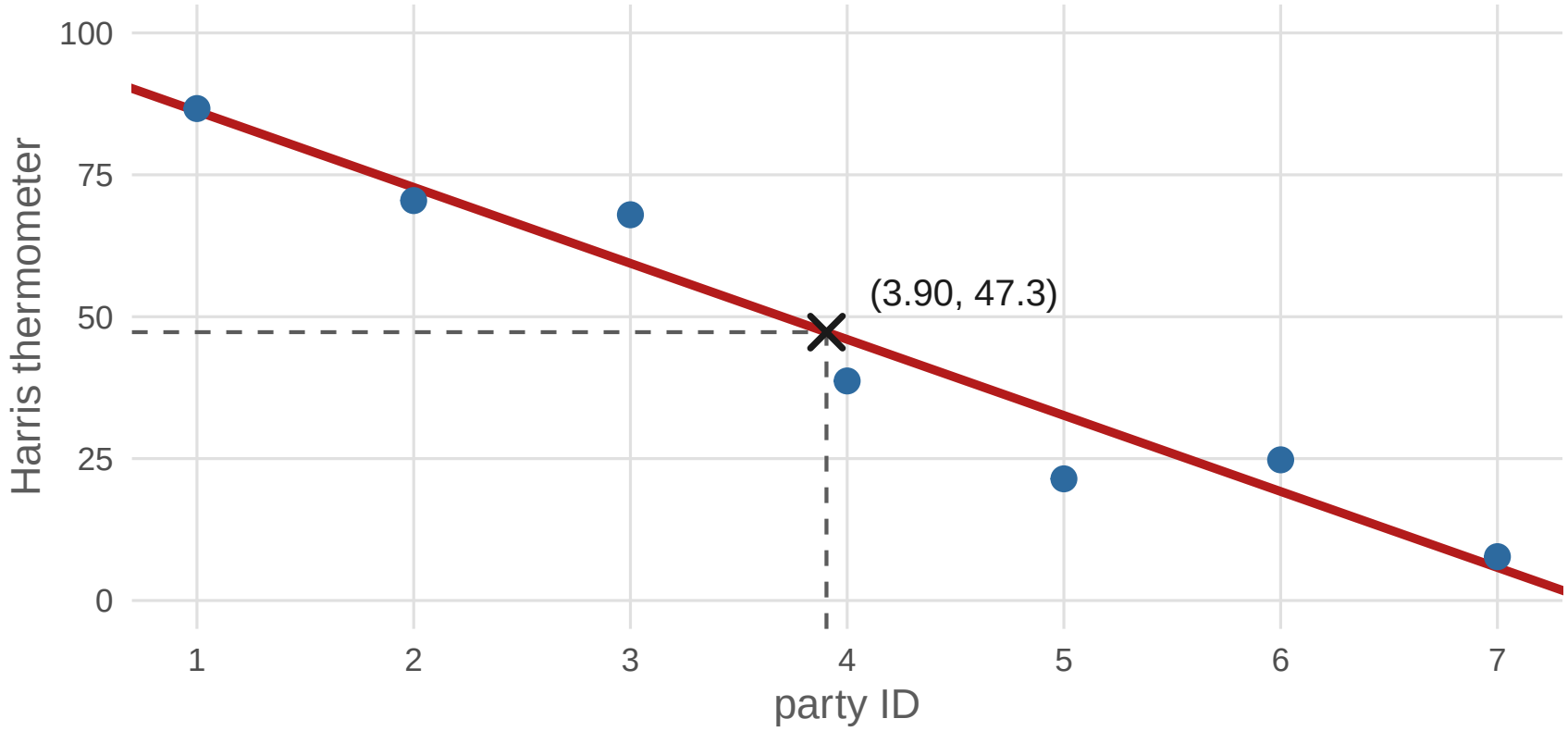
$$= \bar{Y}$$

The point $(\bar{X}, \bar{Y})$ is on the line.

# The means in our example

In our example, $\bar{X} = 3.90$ and $\bar{Y} = 47.3$. The line at $\bar{X}$:

$$99.59 - 13.40 \times 3.90 = 47.3.$$



# Units

## AGE IN YEARS

```
coef(lm(harris ~ age,
        data = anes))
```

```
(Intercept)         age
     41.685       0.111
```

## AGE IN DECADES

```
coef(lm(harris ~ I(age / 10),
        data = anes))
```

```
(Intercept)   I(age/10)
     41.685       1.111
```

$\hat{\beta}_1$ is in units of $Y$ per unit of $X$. Dividing $X$ by 10 multiplies $\hat{\beta}_1$ by 10; $\hat{\beta}_0$ stays the same.

The fitted values do not change: a 50-year-old gets the same $\hat{Y}$ from either line.

# Describing, not explaining

**DESCRIPTIVE**

"is associated with"
"differ by"

*On average, respondents one step more Republican rate Harris 13.4 points lower.*

**CAUSAL**

"the effect of"
"increases" / "reduces"
"leads to"

*Becoming one step more Republican lowers a person's Harris rating by 13.4 points.*

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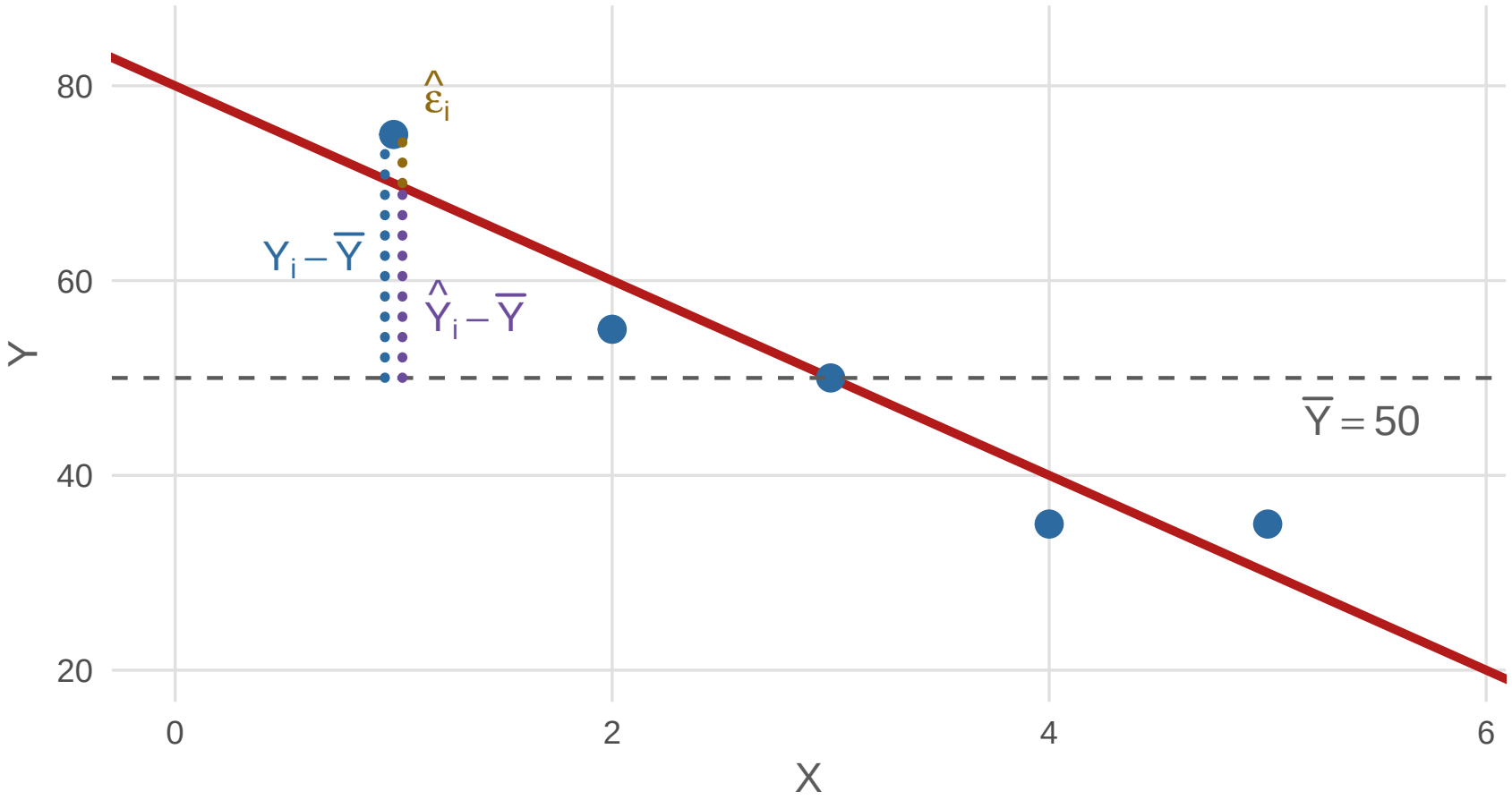
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# One respondent's deviation




$$Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + \hat{\varepsilon}_i$$

# Three sums of squares

**SUMS OF SQUARES**

The **total sum of squares**, the **explained sum of squares** and the **sum of squared residuals**:

$$\text{TSS} = \sum_i (Y_i - \bar{Y})^2, \quad \text{ESS} = \sum_i (\hat{Y}_i - \bar{Y})^2, \quad \text{SSR} = \sum_i \hat{\varepsilon}_i^2.$$

$$\text{TSS} = \text{ESS} + \text{SSR}$$

**COEFFICIENT OF DETERMINATION**

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{SSR}}{\text{TSS}}$$

"$R^2 \times 100\%$ of the variation in $Y$ is accounted for by $X$."
$0 \leq R^2 \leq 1$.

# $R^2$ and the correlation

**BIVARIATE REGRESSION**

$$R^2 = \hat{r}^2$$

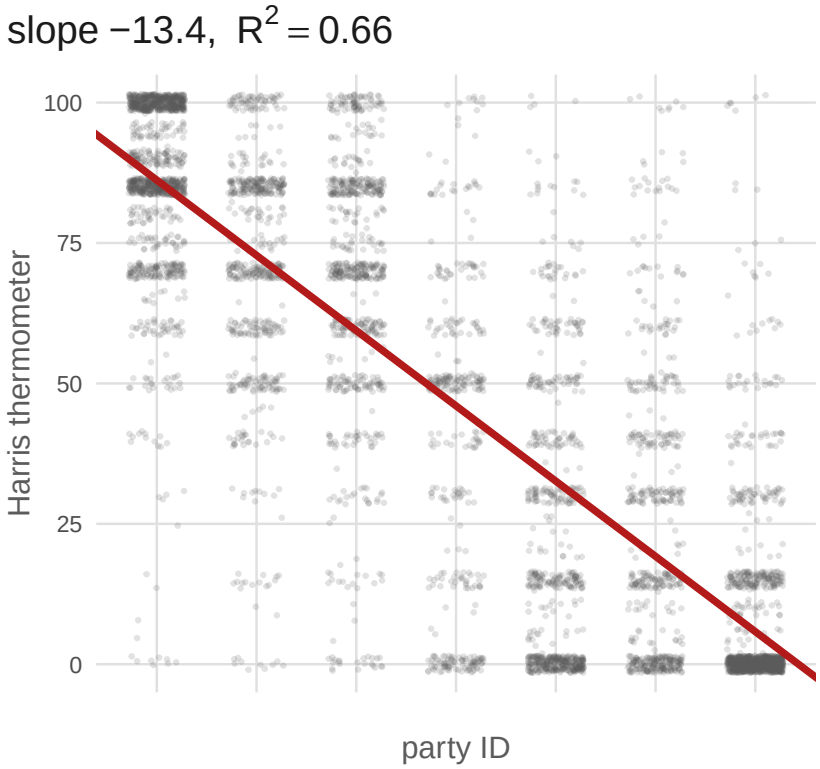
In our running example, the Harris feeling thermometer and `pid`:

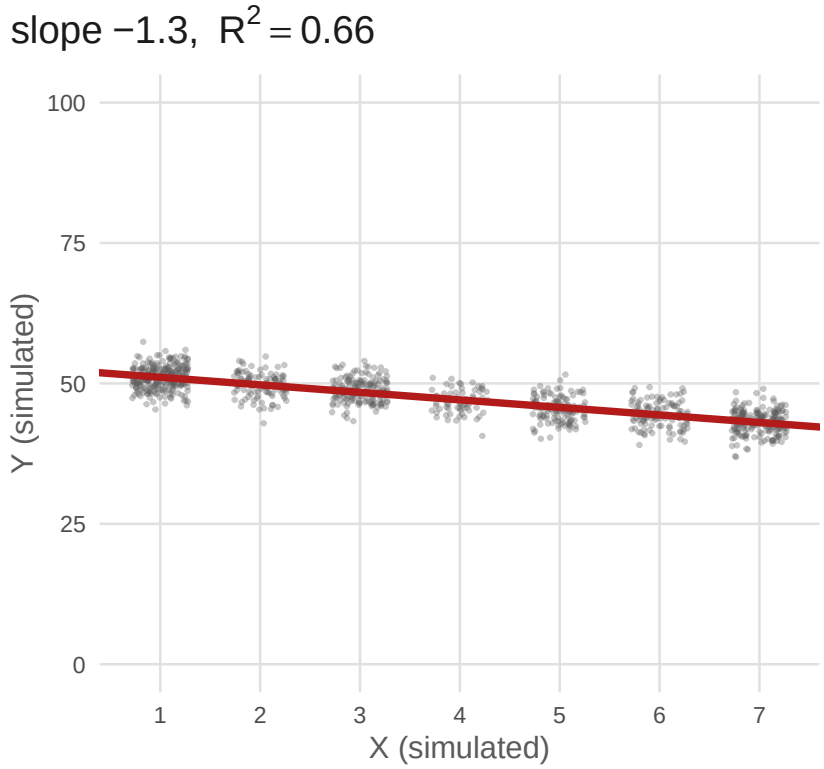
$$\hat{r} = -0.81, \qquad R^2 = (-0.81)^2 = 0.66.$$

66% of the variation in the Harris feeling thermometer is accounted for by `pid`.

Squaring drops the sign: $\hat{r} = 0.81$ would give the same $R^2$.

# Slope size is not fit



# What $R^2$ does not tell you

- Whether a straight line is the right shape for the relationship.
- Whether the slope is substantively large.
- Anything about cause and effect.
- Whether a value is “large” or “small”: there is no universal standard.

# Interpreting `summary()`

```
summary(lm(harris ~ pid, data = anes))
```

```
Call:
lm(formula = harris ~ pid, data = anes)

Residuals:
    Min      1Q  Median      3Q     Max
-86.193  -9.398  -1.193  13.807  94.193

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  99.5907     0.5845   170.4   <2e-16 ***
pid         -13.3977     0.1293  -103.6   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 21.79 on 5465 degrees of freedom
  (54 observations deleted due to missingness)
Multiple R-squared:  0.6627,  Adjusted R-squared:  0.6627
F-statistic: 1.074e+04 on 1 and 5465 DF,  p-value: < 2.2e-16
```

# Roadmap

Relating two variables

Where we are

The difference in means is a regression

From conditional means to a line

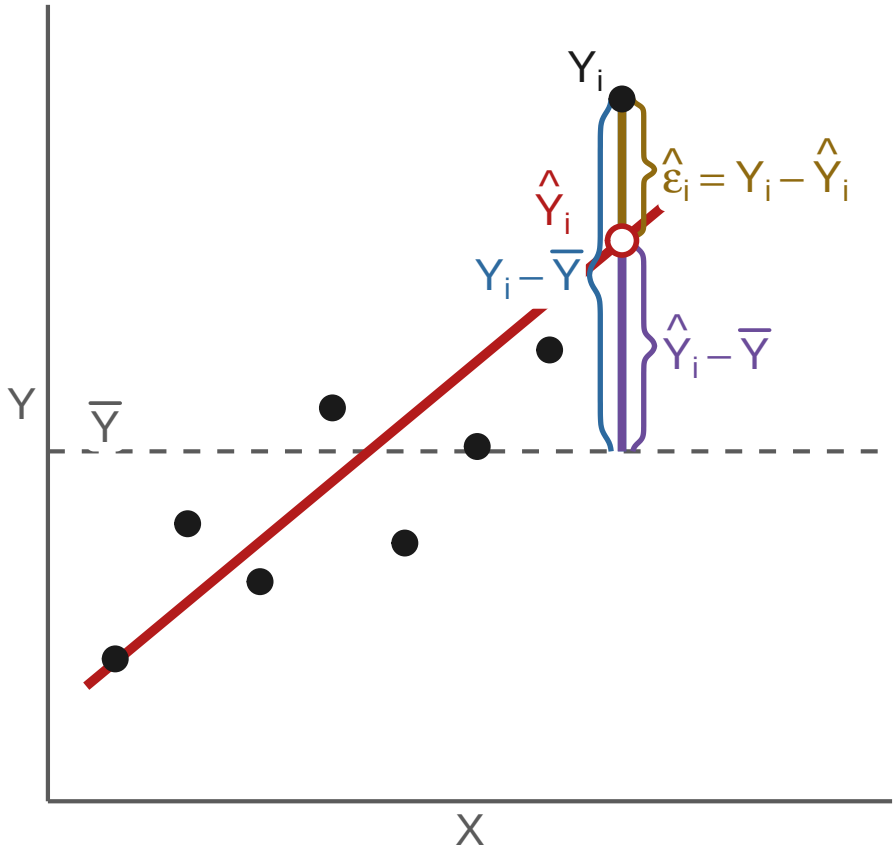
Least squares

Interpreting the line

How well does the line fit?

Wrapping up

# Splitting one deviation




$\bar{Y}$: the mean, the best constant prediction

$Y_i - \bar{Y}$: the total deviation, from the mean

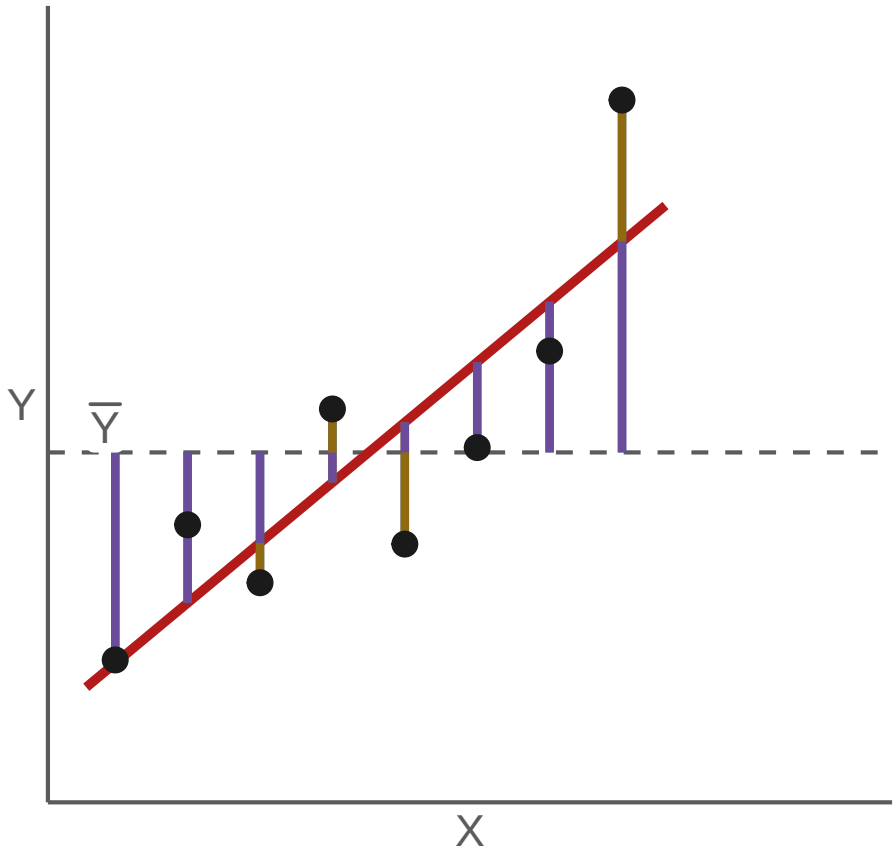
$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$: the fitted value, on the line

$\hat{\varepsilon}_i = Y_i - \hat{Y}_i$: the unexplained deviation, or residual

$\hat{Y}_i - \bar{Y}$: the explained deviation

$$Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + \hat{\varepsilon}_i$$

# Adding up the squares




Total sum of squares:
$\text{TSS} = \sum_i (Y_i - \bar{Y})^2$
The numerator of the variance:
$s_Y^2 = \text{TSS}/(n-1)$.

Sum of squared residuals:
$\text{SSR} = \sum_i (Y_i - \hat{Y}_i)^2$
$= \sum_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$
OLS picks the line with the smallest SSR: the least total squared vertical distance from the points.

Explained (or regression) sum of squares:
$\text{ESS} = \sum_i (\hat{Y}_i - \bar{Y})^2$

$\text{TSS} = \text{ESS} + \text{SSR}$

# What the line accounts for

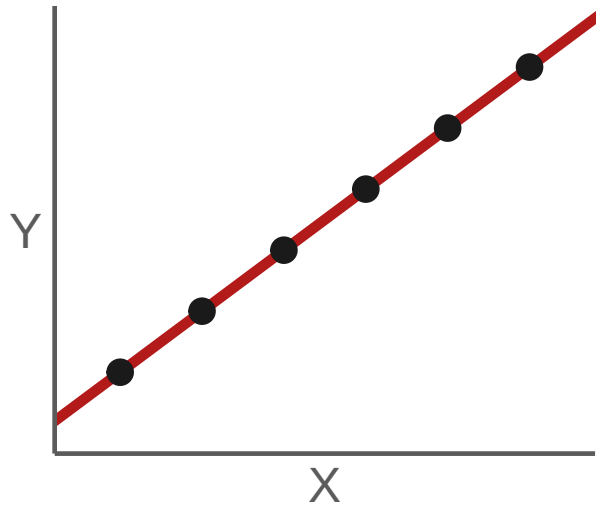
$$R^2 = \frac{\text{TSS} - \text{SSR}}{\text{TSS}} = 1 - \frac{\text{SSR}}{\text{TSS}}$$

$100 \times R^2$: the percent of the variation in $Y$ that $X$ accounts for

If SSR $= 0$, every point sits on the line. Then $R^2 = 1$.

Bivariate regression: $R^2 = \hat{r}^2$

$\hat{\beta}_1$: the average difference in $Y$ between units one unit apart on $X$



# Three best predictions

| Prediction | Criterion | Solution |
|---|---|---|
| One number | $\arg\min_{c} \sum_i (Y_i - c)^2$ | $\bar{Y}$ |
| A line, $X \in \{0, 1\}$ | $\arg\min_{b_0, b_1} \sum_i (Y_i - b_0 - b_1 X_i)^2$ | $\hat{\beta}_0 = \bar{Y}_{X=0}$<br>$\hat{\beta}_1 = \bar{Y}_{X=1} - \bar{Y}_{X=0}$ |
| A line, any $X$ | $\arg\min_{b_0, b_1} \sum_i (Y_i - b_0 - b_1 X_i)^2$ | $\hat{\beta}_1 = \widehat{\text{Cov}}(X, Y)/s_X^2$<br>$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$ |

# Further reading

- Gailmard, *Statistical Modeling and Inference for Social Science*, section 2.3.5.
- Aronow and Miller, *Foundations of Agnostic Statistics*, section 4.1.

These slides draw on teaching materials by Talbot Andrews, Daniel de Kadt, Tom Pepinsky and Teppei Yamamoto.